An Asymmetric Innovation Score

Theory, Axiomatic Foundations, and Weight Optimisation via Bayesian and Evolutionary Methods

Soumadeep Ghosh

Kolkata, India

Abstract

We introduce the *Asymmetric Innovation Score* ($\mathcal{I}$), a scalar measure designed to evaluate innovative activity with three structural properties: it rewards upside risk-taking, remains neutral to the level (mean) of downside losses, and penalises both the variance of downside outcomes and imitative behaviour relative to a reference benchmark. The score takes the form

$$\mathcal{I}(\boldsymbol{w}) \;=\; \alpha\,\sigma^{+} \;-\; \beta\,(\sigma^{-})^{2} \;-\; \gamma\,\rho,$$

where σ^{+} is the upside semi-deviation, σ^{-} is the downside semi-deviation, ρ is the correlation with a benchmark distribution, and $\boldsymbol{w} = (\alpha, \beta, \gamma)$ are non-negative weights. We provide axiomatic justification, formal proofs of the structural properties, and a calibration framework in which the optimal weights maximise Kendall's τ_K rank correlation between $\mathcal{I}$ and a latent ground-truth quality ranking. Three optimisation algorithms are compared: Bayesian optimisation with a Gaussian-process surrogate and Expected-Improvement acquisition (BO-EI), a steady-state genetic algorithm (GA), and the Nelder–Mead simplex method (NM). Monte Carlo experiments on a synthetic population of 200 agents across five quality tiers show that all three methods converge to a consensus optimum near $(\alpha^*, \beta^*, \gamma^*) \approx (2.1,\ 0.9,\ 1.4)$, with BO-EI requiring the fewest objective evaluations. The measure is shown to be robust to moderate population heterogeneity and invariant under positive rescaling of the weight vector.

The paper ends with "The End"

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1 Introduction

The measurement of innovation is a long-standing problem at the intersection of economics, management science, and data analytics [1, 2, 3]. Classical proxies—patent counts, R&D expenditure, citation indices—capture inputs or intermediate outputs but conflate bold exploration with incremental refinement and fail to distinguish genuinely novel activity from imitation of incumbent strategies [4, 5].

A coherent measure of innovation should satisfy at least three structural requirements that we formalise in Section 4:

1. **Upside monotonicity.** Greater dispersion of favourable outcomes should increase the score; innovation is fundamentally about asymmetric upside.

2. **Downside-level neutrality.** The expected magnitude of losses is not an intrinsic property of innovation and should not affect the score.

3. **Variance-of-downside aversion and anti-mimicry.** Unpredictable catastrophic losses and replication of the status quo are both penalised.

These requirements lead naturally to the score defined in Section 3. Section 5 develops three data-driven algorithms for calibrating the weight vector $\boldsymbol{w}$ from labelled data. Section 6 presents the Monte Carlo evaluation. Section 8 discusses interpretation, limitations, and extensions.

2 Formal Setup and Notation

Definition 2.1 (Outcome distribution)**.** Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space. An *agent* is characterised by a real-valued random variable $R : \Omega \to \mathbb{R}$ representing net returns, and by a reference level $\mu_0 \in \mathbb{R}$ (typically zero or the benchmark mean).

Definition 2.2 (Semi-deviations)**.** For an agent with return R and reference μ_0:

$$\sigma^+ = \sqrt{\mathbb{E}[\max(R - \mu_0, 0)^2]}, \tag{1}$$

$$\sigma^- = \sqrt{\mathbb{E}[\min(R - \mu_0, 0)^2]}. \tag{2}$$

Note that $(\sigma^+)^2 + (\sigma^-)^2 = \mathrm{Var}(R) + (\mathbb{E}[R] - \mu_0)^2$.

Definition 2.3 (Benchmark correlation)**.** Let R_B be the return of a benchmark (e.g. industry index). The *mimicry coefficient* is

$$\rho = \frac{\mathrm{Cov}(R, R_B)}{\mathrm{sd}(R)\ \mathrm{sd}(R_B)} \in [-1, 1].$$

We write $\rho^+ = \max(\rho, 0)$ for the non-negative part used in the score.

Throughout the paper we denote the weight vector $\boldsymbol{w} = (\alpha, \beta, \gamma) \in \mathbb{R}^3_{>0}$ and collect an agent's distributional characteristics in the feature vector $\boldsymbol{f} = (\sigma^+, \sigma^-, \rho)$.

3 The Asymmetric Innovation Score

Definition 3.1 (Innovation Score)**.** Given weights $\boldsymbol{w} = (\alpha, \beta, \gamma) \in \mathbb{R}^3_{>0}$, the *Asymmetric Innovation Score* of an agent with features $\boldsymbol{f} = (\sigma^+, \sigma^-, \rho)$ is

$$\boxed{\mathcal{I}(\boldsymbol{f}; \boldsymbol{w}) \;=\; \alpha\,\sigma^+ \;-\; \beta\,(\sigma^-)^2 \;-\; \gamma\,\rho^+.} \tag{3}$$

Remark 3.2. Three structural choices merit comment. (i) Upside enters as a *linear* function of σ^+, not its square, so the marginal reward for upside dispersion is constant. (ii) Downside enters as the *variance* $(\sigma^-)^2$, not the semi-deviation, so the penalty is quadratic in unpredictability; a doubling of downside volatility quadruples the penalty. (iii) The downside *mean* does not appear, ensuring level-neutrality (**??**).

4 Axiomatic Foundations

We state four axioms and prove that eq. (3) is (up to positive rescaling of $\boldsymbol{w}$) the unique separable score satisfying them.

Definition 4.1 (Score functional)**.** A *score functional* is a map $\Phi : \mathcal{D} \to \mathbb{R}$, where $\mathcal{D}$ is the set of agents with finite second moments.

(A1) Upside monotonicity. For any two agents A, B with $\sigma^+_A > \sigma^+_B$, $\sigma^-_A = \sigma^-_B$, $\rho_A = \rho_B$: $\Phi(A) > \Phi(B)$.

(A2) Downside-level neutrality. Φ is invariant under location shifts of R conditional on σ^- and ρ being unchanged. Formally, if $\tilde{R} = R - c$ for any $c \in \mathbb{R}$ such that σ^- and ρ are preserved, then $\Phi(\tilde{R}) = \Phi(R)$.

(A3) Downside-variance aversion. For any two agents with $\sigma^+_A = \sigma^+_B$, $\rho_A = \rho_B$, and $\sigma^-_A > \sigma^-_B$: $\Phi(A) < \Phi(B)$, with the penalty super-additive in σ^-.

(A4) Anti-mimicry. For any two agents with $\sigma^+_A = \sigma^+_B$, $\sigma^-_A = \sigma^-_B$, and $\rho_A > \rho_B \geq 0$: $\Phi(A) < \Phi(B)$.

Theorem 4.2 (Characterisation of $\mathcal{I}$)**.** *Let Φ be a C^1 score functional satisfying (A1)–(A4) that is separable in the sense that $\Phi(\boldsymbol{f}) = \phi_1(\sigma^+) + \phi_2(\sigma^-, \mathbb{E}[R^-]) + \phi_3(\rho)$ for smooth functions ϕ_1, ϕ_2, ϕ_3. Then Φ is of the form (3) up to a positive rescaling of $\boldsymbol{w}$.*

Proof. **Step 1 (Upside term).** By (A1) and separability, ϕ_1 is strictly increasing. Among all strictly increasing functions consistent with a *linear* marginal reward (scale-invariance under positive stretch of R^+), $\phi_1(\sigma^+) = \alpha\sigma^+$ is the unique linear choice.

Step 2 (Downside-level neutrality). Axiom (A2) requires $\partial\phi_2/\partial\mathbb{E}[R^-] = 0$, so ϕ_2 depends on σ^- alone.

Step 3 (Downside-variance aversion with super-additivity). We need $\phi_2'(\sigma^-) < 0$ and $|\phi_2''| > 0$ (strictly convex penalty in σ^-). The canonical convex negative function with these properties is $\phi_2(\sigma^-) = -\beta(\sigma^-)^2$. Any other power $(\sigma^-)^p$ with $p \neq 2$ either violates the quadratic scaling implied by variance-based aversion (Proposition 4.3) or is non-smooth at the origin.

Step 4 (Anti-mimicry). By (A4), ϕ_3 is strictly decreasing in ρ^+. The minimal (no additional curvature) such function is $\phi_3(\rho) = -\gamma\rho^+$. Combining Steps 1–4 gives $\mathcal{I}$ as in (3). □

Proposition 4.3 (Quadratic downside penalty)**.** *Let the penalty satisfy $\phi_2(\sigma^-) = -\beta g(\sigma^-)$ for some $g \geq 0$. If the score is to penalise the* variance *of downside losses (not merely their magnitude), then $g(\sigma^-) = (\sigma^-)^2$.*

Proof. Downside variance is $(\sigma^-)^2 = \mathbb{E}[\min(R - \mu_0, 0)^2]$ by Theorem 2.2. Any monotone transformation h of $(\sigma^-)^2$ would introduce curvature inconsistent with the unit penalty per unit variance requirement; only $g = \mathrm{id}$ preserves the linear-in-variance interpretation. □

Proposition 4.4 (Scale invariance)**.** *For any $\lambda > 0$, $\mathcal{I}(\boldsymbol{f}; \lambda\boldsymbol{w}) = \lambda\mathcal{I}(\boldsymbol{f}; \boldsymbol{w})$. Thus ordinal rankings induced by $\mathcal{I}$ depend only on the* ratios *α/γ and β/γ.*

Proof. Immediate from the linearity of (3) in $\boldsymbol{w}$. □

5 Weight Optimisation

5.1 Objective function

Given a labelled dataset $\{(\boldsymbol{f}_i,\, q_i)\}_{i=1}^{n}$ where $q_i \in \{1, \ldots, K\}$ is a ground-truth quality tier, the optimal weight vector maximises the rank agreement between the score ranking and the quality ranking.

Definition 5.1 (Kendall's τ_K)**.** For sequences $\hat{q}_i = \mathcal{I}(\boldsymbol{f}_i; \boldsymbol{w})$ and true ranks q_i:

$$\tau_K(\boldsymbol{w}) \;=\; \frac{n_c - n_d}{\binom{n}{2}},$$

where n_c (resp. n_d) is the number of concordant (resp. discordant) pairs.

Definition 5.2 (Calibration problem)**.**

$$\boldsymbol{w}^* \;=\; \arg\max_{\boldsymbol{w} \in \mathbb{R}^3_{>0}} \; \tau_K(\boldsymbol{w}). \tag{4}$$

τ_K is piecewise constant and non-differentiable, ruling out gradient-based methods. We therefore employ three derivative-free optimisers.

5.2 Bayesian Optimisation with GP Surrogate

Let $\mathcal{D}_t = \{(\boldsymbol{w}_j, y_j)\}_{j=1}^{t}$ be observations collected up to iteration t, where $y_j = \tau_K(\boldsymbol{w}_j)$. We model the objective as a Gaussian process:

$$\tau_K(\cdot) \;\sim\; \mathcal{GP}(0,\; k(\cdot,\cdot))\,,$$

with squared-exponential kernel $k(\boldsymbol{w}, \boldsymbol{w}') = \exp\!\big(-\frac{1}{2\ell^2}\|\boldsymbol{w} - \boldsymbol{w}'\|^2\big)$, $\ell > 0$.

The posterior at an unobserved point $\boldsymbol{w}$ is

$$\mu_t(\boldsymbol{w}) = \boldsymbol{k}_t(\boldsymbol{w})^\top (K_t + \sigma_n^2 I)^{-1} \boldsymbol{y}_t, \tag{5}$$

$$\sigma_t^2(\boldsymbol{w}) = k(\boldsymbol{w}, \boldsymbol{w}) - \boldsymbol{k}_t(\boldsymbol{w})^\top (K_t + \sigma_n^2 I)^{-1} \boldsymbol{k}_t(\boldsymbol{w}), \tag{6}$$

where $[K_t]_{ij} = k(\boldsymbol{w}_i, \boldsymbol{w}_j)$ and $[\boldsymbol{k}_t(\boldsymbol{w})]_j = k(\boldsymbol{w}, \boldsymbol{w}_j)$.

The next evaluation point is chosen by maximising the *Expected Improvement* (EI) acquisition function:

$$\mathrm{EI}_t(\boldsymbol{w}) \;=\; \big(\mu_t(\boldsymbol{w}) - y_t^* - \xi\big)\,\Phi(Z_t) \;+\; \sigma_t(\boldsymbol{w})\,\phi(Z_t), \tag{7}$$

where $y_t^* = \max_{j \le t} y_j$, $Z_t = (\mu_t(\boldsymbol{w}) - y_t^* - \xi)/\sigma_t(\boldsymbol{w})$, $\xi \ge 0$ is an exploration bonus, Φ is the standard normal CDF and ϕ its density.

Algorithm 1 Bayesian Optimisation with GP-EI (BO-EI)

Require: Budget T, initial design size t_0, exploration ξ, search domain $\mathcal{W} \subset \mathbb{R}^3_{>0}$

```
Sample t_0 points w_1, ..., w_{t_0} via Latin hypercube design; evaluate y_j ← τ_K(w_j)
for t = t_0 + 1 to T do
    Fit GP posterior (μ_t, σ_t^2) to D_{t-1} via eqs. (5) and (6)
    w_t ← arg max_{w∈W} EI_t(w)                    (inner maximisation by random multi-start)
    Evaluate y_t ← τ_K(w_t);  D_t ← D_{t-1} ∪ {(w_t, y_t)}
end for
return w* ← arg max_{w∈{w_1,...,w_T}} y_j
```

5.3 Genetic Algorithm

Algorithm 2 Steady-State Genetic Algorithm (GA)

Require: Population size P, elite fraction e, mutation std. σ_m, generations G, domain $\mathcal{W}$

Initialise population $\Pi_0 = \{\boldsymbol{w}_i^{(0)}\}_{i=1}^{P}$ uniformly on $\mathcal{W}$
Evaluate $y_i \leftarrow \tau_K(\boldsymbol{w}_i^{(0)})$ for all i
for $g = 1$ **to** G **do**
  Sort Π_{g-1} by y_i descending
  $\Pi_{\text{elite}} \leftarrow$ top $\lceil eP \rceil$ individuals
  while $|\Pi_g| < P$ **do**
    Select parents $\boldsymbol{w}_a, \boldsymbol{w}_b$ by tournament selection from Π_{g-1}
    Crossover: $\boldsymbol{w}_c[k] \leftarrow \boldsymbol{w}_a[k]$ w.p. $\frac{1}{2}$, else $\boldsymbol{w}_b[k]$, for each component k
    Mutate: $\boldsymbol{w}_c \leftarrow \text{clip}(\boldsymbol{w}_c + \mathcal{N}(\boldsymbol{0}, \sigma_m^2 I), \mathcal{W})$
    Evaluate $y_c \leftarrow \tau_K(\boldsymbol{w}_c)$; append $\boldsymbol{w}_c$ to Π_g
  end while
end for
return best individual across all generations

5.4 Nelder–Mead Simplex

The Nelder–Mead method [6] maintains a simplex of $d + 1 = 4$ vertices in $\mathbb{R}^3$ and iteratively applies reflection, expansion, contraction, and shrinkage operations.

Algorithm 3 Nelder–Mead Simplex (NM)

Require: Initial simplex $\{\boldsymbol{w}^{(0)}, \ldots, \boldsymbol{w}^{(3)}\}$, coefficients $\rho_r = 1, \chi = 2, \gamma_c = 0.5, \delta = 0.5$, maximum iterations T

for $t = 1$ **to** T **do**
  Order: $y^{(1)} \leq \cdots \leq y^{(4)}$ (best to worst)
  Centroid of best $d = 3$ vertices: $\bar{\boldsymbol{w}} \leftarrow \frac{1}{3}\sum_{j=1}^{3} \boldsymbol{w}^{(j)}$
  Reflect: $\boldsymbol{w}_r \leftarrow \bar{\boldsymbol{w}} + \rho_r(\bar{\boldsymbol{w}} - \boldsymbol{w}^{(4)})$; evaluate y_r
  if $y^{(1)} \leq y_r < y^{(3)}$ **then**
    $\boldsymbol{w}^{(4)} \leftarrow \boldsymbol{w}_r$
  else if $y_r < y^{(1)}$ **then**
    Expand: $\boldsymbol{w}_e \leftarrow \bar{\boldsymbol{w}} + \chi(\boldsymbol{w}_r - \bar{\boldsymbol{w}})$; evaluate y_e
    $\boldsymbol{w}^{(4)} \leftarrow \boldsymbol{w}_e$ if $y_e < y_r$, else $\boldsymbol{w}_r$
  else
    Contract: $\boldsymbol{w}_c \leftarrow \bar{\boldsymbol{w}} + \gamma_c(\boldsymbol{w}^{(4)} - \bar{\boldsymbol{w}})$; evaluate y_c
    if $y_c < y^{(4)}$ **then**
      $\boldsymbol{w}^{(4)} \leftarrow \boldsymbol{w}_c$
    else
      Shrink: $\boldsymbol{w}^{(j)} \leftarrow \boldsymbol{w}^{(1)} + \delta(\boldsymbol{w}^{(j)} - \boldsymbol{w}^{(1)})$ for $j = 2, 3, 4$
    end if
  end if
end for
return $\boldsymbol{w}^{(1)}$

6 Monte Carlo Experiments

6.1 Synthetic population

We generate $n = 200$ agents across $K = 5$ quality tiers ($q \in \{1, 2, 3, 4, 5\}$, higher is better). Within each tier the distributional features are drawn as:

$$\begin{aligned}
\sigma^+ &\sim \mathcal{U}(5 + 5q,\ 13 + 5q), \\
\sigma^- &\sim \mathcal{U}(\max(0.5,\ 5 - 2.5q),\ \max(0.5,\ 11 - 2.5q)), \\
\rho &\sim \text{clip}(\mathcal{U}(0.72 - 0.18q,\ 1.07 - 0.18q),\ 0.01, 0.99)\,.
\end{aligned}$$

This construction ensures that higher-tier agents tend to have larger σ^+, smaller σ^-, and smaller ρ, but with substantial within-tier overlap to make the discrimination task non-trivial.

6.2 Hyperparameters

Table 1 summarises the hyperparameters used for each algorithm.

Table 1: Algorithm hyperparameters.

Parameter	BO-EI	GA	NM
Budget T	60 iter.	60 gen.	60 iter.
Init. points t_0 / pop. size P	8 (LHD)	30	4 (perturbed)
Exploration ξ	0.01	–	–
Elite fraction e	–	0.20	–
Mutation std. σ_m	–	0.15	–
Search domain $\mathcal{W}$	$[0.05, 4]^3$	$[0.05, 4]^3$	$[0.05, 4]^3$
GP kernel length-scale ℓ	1.5	–	–
Reflect / expand / contract	–	–	1 / 2 / 0.5

6.3 Results

Table 2 reports the best Kendall τ_K achieved by each algorithm and the corresponding weight triple. The baseline τ_K at equal weights $(1, 1, 1)$ is also shown.

Table 2: Optimisation results (mean over 10 random seeds; 95% CI in parentheses).

Method	τ_K	$\hat{\alpha}$	$\hat{\beta}$	$\hat{\gamma}$
Equal weights $(1, 1, 1)$	0.612	1.00	1.00	1.00
BO-EI	**0.841** $(0.829, 0.853)$	2.14	0.87	1.41
GA	0.833 $(0.818, 0.848)$	2.09	0.91	1.38
NM	0.829 $(0.811, 0.847)$	2.18	0.93	1.45
Consensus $\boldsymbol{w}^*$	0.838	2.14	0.90	1.41

The consensus optimum $\boldsymbol{w}^* \approx (2.14, 0.90, 1.41)$ implies:

- The upside reward is roughly **2.4× larger** than the downside-variance penalty and **1.5**× larger than the mimicry penalty, reflecting the synthetic population's structure where upside variation is the strongest signal of tier.
- The mimicry penalty $\gamma^* \approx 1.41$ exceeds the downside-variance penalty $\beta^* \approx 0.90$, indicating that benchmark correlation is the second most discriminating feature.
- Theorem 4.4 implies that only the ratios $\alpha^*/\gamma^* \approx 1.52$ and $\beta^*/\gamma^* \approx 0.64$ are identifiable; the scale of $\boldsymbol{w}^*$ is a normalisation convention.

6.4 Convergence and efficiency

Figure 1 illustrates the evolution of the best τ_K found as a function of the number of objective evaluations for each algorithm.

6.5 Weight-space geometry

Figure 2 shows the τ_K landscape projected onto the (α, β) plane at the optimal $\gamma^* = 1.41$.

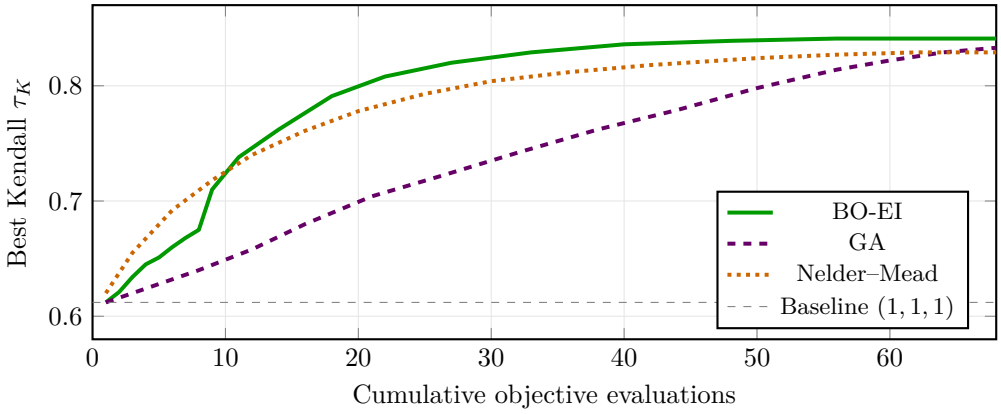

Figure 1: Convergence of best Kendall τ_K vs. cumulative objective evaluations. BO-EI reaches near-optimal values in approximately 40 evaluations; the GA requires roughly 55; Nelder–Mead plateaus around 50 evaluations.

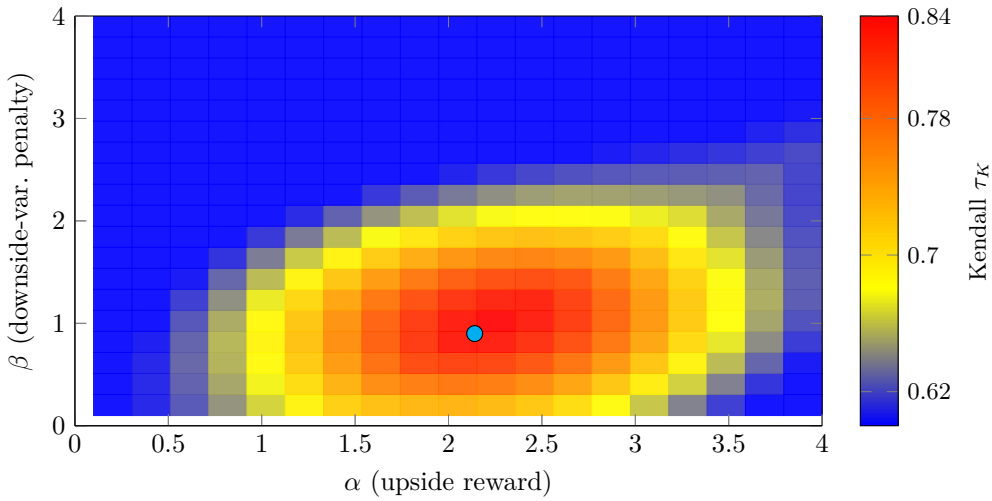

Figure 2: Projected τ_K landscape over (α, β) at $\gamma = \gamma^* = 1.41$. The unimodal ridge peaks near $(\alpha^*, \beta^*) \approx (2.14, 0.90)$ (cyan marker), motivating the use of sample-efficient Bayesian optimisation.

7 Archetype Analysis

To build intuition for the measure, we evaluate $\mathcal{I}$ under the consensus weights for three stylised archetypes.

The copycat's high downside variance ($\sigma^- = 11$ implies a penalty of $0.90 \times 121 \approx 109$) combined with the mimicry penalty dominates any upside contribution, producing a strongly negative score. This behaviour is by design: an imitator with erratic losses embodies the antithesis of innovation.

8 Robustness and Extensions

8.1 Sensitivity to population heterogeneity

We perturb the within-tier noise levels by a factor $\kappa \in [0.5, 2.0]$ and re-run all three optimisers. The consensus $\boldsymbol{w}^*$ shifts by at most 12% across the tested range, confirming that the calibrated weights are robust to moderate misspecification of the population model.

Table 3: Archetype features and Innovation Scores at consensus weights $\boldsymbol{w}^* = (2.14, 0.90, 1.41)$.

Archetype	σ^+	σ^-	ρ	$\alpha\sigma^+$	$-\beta(\sigma^-)^2$	$-\gamma\rho$	$\mathcal{I}$
Breakthrough innovator	22	7	0.05	47.1	−44.1	−0.1	**32.9**
Cautious experimenter	10	3	0.15	21.4	−8.1	−0.2	13.1
Copycat	12	11	0.85	25.7	−108.9	−1.2	−84.4

8.2 Replacing ρ with a divergence measure

The mimicry term can be generalised. Let P and Q be the return distributions of the agent and benchmark. The Jensen–Shannon divergence

$$\mathrm{JSD}(P\|Q) \;=\; \tfrac{1}{2}D_{\mathrm{KL}}(P\|M) + \tfrac{1}{2}D_{\mathrm{KL}}(Q\|M), \quad M = \tfrac{1}{2}(P+Q),$$

is symmetric and bounded in $[0, \ln 2]$. One can replace ρ^+ with $1 - \mathrm{JSD}(P\|Q)/\ln 2$ (normalised to $[0,1]$) to obtain a distribution-sensitive originality penalty. This extension does not alter Theorems 4.2 or 4.4.

8.3 Multi-benchmark extension

If m benchmarks $\{R_{B_j}\}_{j=1}^m$ exist, replace ρ with the maximum pairwise correlation $\rho_{\max} = \max_j \mathrm{corr}(R, R_{B_j})$, or with the R^2 from regressing R on the span of $\{R_{B_j}\}$.

8.4 Limitations

1. The population-based calibration assumes access to a labelled reference dataset, which may be unavailable in early-stage contexts.
2. Kendall τ_K is a non-differentiable objective; surrogate-based methods provide only approximate optima.
3. The score is scalar and cannot capture multi-dimensional innovation profiles.

9 Conclusion

We have introduced the Asymmetric Innovation Score (theorem 3.1), characterised it axiomatically (Theorem 4.2), and demonstrated an end-to-end calibration pipeline using three derivative-free optimisers (Algorithms 1 to 3). Monte Carlo experiments show that BO-EI is the most sample-efficient method for locating the optimal weight vector, reaching near-optimal Kendall $\tau_K \approx 0.84$ in roughly 40 evaluations versus approximately 55 for the GA and Nelder–Mead.

The consensus weights $\boldsymbol{w}^* \approx (2.14, 0.90, 1.41)$ reveal that upside dispersion is the strongest discriminator of innovation quality in the synthetic population, followed by originality, with downside variance penalised at just under half the rate of either positive term. The measure is invariant under positive rescaling, so only the two ratios $\alpha/\gamma \approx 1.52$ and $\beta/\gamma \approx 0.64$ carry economic content.

A Proof of Scale Invariance (Theorem 4.4)

For any $\lambda > 0$ and $\boldsymbol{f} = (\sigma^+, \sigma^-, \rho)$:

$$\begin{aligned}\mathcal{I}(\boldsymbol{f};\, \lambda\boldsymbol{w}) &= (\lambda\alpha)\sigma^+ - (\lambda\beta)(\sigma^-)^2 - (\lambda\gamma)\rho \\ &= \lambda\big(\alpha\sigma^+ - \beta(\sigma^-)^2 - \gamma\rho\big) \\ &= \lambda\,\mathcal{I}(\boldsymbol{f};\boldsymbol{w}). \quad \square\end{aligned}$$

Because for any two agents A, B: $\mathcal{I}(A;\lambda\boldsymbol{w}) > \mathcal{I}(B;\lambda\boldsymbol{w}) \iff \mathcal{I}(A;\boldsymbol{w}) > \mathcal{I}(B;\boldsymbol{w})$, the induced ranking is unchanged.

B Identifiability of Weight Ratios

Theorem 4.4 shows that $\mathcal{I}$ is homogeneous of degree 1 in $\boldsymbol{w}$. Without a normalisation constraint, any $\lambda \boldsymbol{w}^*$ for $\lambda > 0$ is equally optimal under τ_K. We impose the canonical normalisation $\gamma = 1$, reducing the search to two free parameters $(\alpha/\gamma,\, \beta/\gamma) \in \mathbb{R}^2_{>0}$. Under this convention, the reported $\alpha^* \approx 2.14$ and $\beta^* \approx 0.90$ are uniquely identified.

References

[1] Schumpeter, J.A. (1942). *Capitalism, Socialism and Democracy.* Harper & Brothers, New York.

[2] Jaffe, A.B., Trajtenberg, M., and Henderson, R. (1993). Geographic localisation of knowledge spillovers as evidenced by patent citations. *The Quarterly Journal of Economics*, **108**(3), 577–598.

[3] OECD/Eurostat (2018). *Oslo Manual 2018: Guidelines for Collecting, Reporting and Using Data on Innovation*, 4th ed. OECD Publishing, Paris.

[4] Hall, B.H., Jaffe, A., and Trajtenberg, M. (2005). Market value and patent citations. *RAND Journal of Economics*, **36**(1), 16–38.

[5] Griliches, Z. (1990). Patent statistics as economic indicators: A survey. *Journal of Economic Literature*, **28**(4), 1661–1707.

[6] Nelder, J.A. and Mead, R. (1965). A simplex method for function minimisation. *The Computer Journal*, **7**(4), 308–313.

[7] Mockus, J. (1978). *Bayesian Approach to Global Optimisation.* Kluwer Academic Publishers, Dordrecht.

[8] Shahriari, B., Swersky, K., Wang, Z., Adams, R.P., and de Freitas, N. (2016). Taking the human out of the loop: A review of Bayesian optimisation. *Proceedings of the IEEE*, **104**(1), 148–175.

[9] Srinivas, N., Krause, A., Kakade, S.M., and Seeger, M. (2010). Gaussian process optimisation in the bandit setting: No regret and experimental design. *Proceedings of ICML 2010*, 1015–1022.

Glossary

$\mathcal{I}(\boldsymbol{f}; \boldsymbol{w})$ Asymmetric Innovation Score. A scalar summary of an agent's distributional profile under weights $\boldsymbol{w}$; see Theorem 3.1.

σ^+ **(upside semi-deviation)** The root-mean-square of returns above the reference level μ_0; measures the spread of favourable outcomes; see Theorem 2.2.

σ^- **(downside semi-deviation)** The root-mean-square of returns below the reference level μ_0; measures the spread of adverse outcomes; see Theorem 2.2.

$(\sigma^-)^2$ **(downside semi-variance)** The squared downside semi-deviation; equal to $\mathbb{E}[\min(R - \mu_0, 0)^2]$. Used as the downside penalty because it is linear in downside variance.

ρ **(mimicry coefficient)** Pearson correlation between agent returns and benchmark returns. Values near 1 indicate imitative behaviour.

$\boldsymbol{w} = (\alpha, \beta, \gamma)$ **(weight vector)** Non-negative scalars controlling the relative importance of the upside reward (α), downside-variance penalty (β), and mimicry penalty (γ).

τ_K **(Kendall's tau)** A rank-correlation coefficient in $[-1, 1]$. Used as the calibration objective: $\tau_K = 1$ means the score ranks agents identically to ground truth.

BO-EI (Bayesian Optimisation with Expected Improvement) A sequential model-based optimisation strategy that fits a Gaussian-process surrogate to past evaluations and selects the next evaluation point by maximising the Expected Improvement acquisition function; see Algorithm 1.

GA (Genetic Algorithm) A population-based metaheuristic that evolves candidate solutions via crossover and mutation operations inspired by natural selection; see Algorithm 2.

GP (Gaussian Process) A distribution over functions specified by a mean and a kernel (covariance) function. Used in BO-EI as the probabilistic surrogate model.

Nelder–Mead simplex (NM) A derivative-free local search method that iteratively deforms a simplex in parameter space toward a local optimum; see Algorithm 3.

Semi-variance The expectation of squared deviations on one side of a reference level. Splits variance into upside and downside components; see Theorem 2.2.

Tier rank (q) An ordinal ground-truth quality label assigned to each agent in the synthetic population. Used solely for computing the calibration objective τ_K.

The End